

## 1 L1 Introduction

**Computer Vision (CV):** 用计算机理解 image/video 的 meaning, 即 Marr: know what is where by looking. 图像来自 sensing device, CV 是 graphics 的 inverse problem: graphics model  $\rightarrow$  image, vision image  $\rightarrow$  world/model. **Understand image:** objects/scene, 3D/material properties, actions, relations, 可回答关于图像的问题. **Applications:** measurement (shape/motion capture, 3D urban modeling, camera ruler, photo tourism), recognition (face/OCR/fine-grained/action), reconstruction, robotics, AR/VR, medical, autonomous driving. **Core difficulties:** inverse problem is ill-posed; projection loses depth; appearance changes with illumination/viewpoint/occlusion/deformation/background; semantic gap from pixels to meaning. **Task modes:** measurement (shape/motion/3D), recognition (semantic content), manipulation/generation (inpainting, style/domain transfer, deepfake). Progress is dataset-driven: larger/richer data enables captioning, interaction, egocentric/video and multimodal tasks.

## 2 L2 Image Basics & Filters

### 2.1 Digital Image

Digital camera: sensor array converts photons to electrons. Image formation includes sampling continuous image plane and quantizing intensity. Grayscale image  $I: \mathbb{R}^2 \rightarrow \mathbb{R}$ ; digital image  $I \in \mathbb{R}^{H \times W \times C}$ , usually uint8 [0,255]. Resolution is spatial sampling density. Low sampling causes aliasing; quantization causes intensity error. **Image as function:** transformation can be pointwise/intensity, e.g.  $g(x,y) = f(x,y) + 50$ , geometric  $g(x,y) = f(x,-y)$ , threshold/mask  $g = f \cdot \mathbf{1}[f < 250]$ . **Color:** perceived property from light/surface/visual system. RGB is linear primary weights; HSV separates hue/saturation/value, useful when color appearance and brightness need separate handling. Histogram counts intensities/color bins; useful for contrast, thresholding, matching, normalization. **Noise:** salt-and-pepper = random black/white spikes; impulse = sparse white spikes; Gaussian = iid normal intensity perturbation. Mean/Gaussian smoothing assumes neighboring pixels are similar and noise is independent, so averaging suppresses noise but blurs edges.

### 2.2 Correlation, Convolution, Filtering

**Correlation:**  $G = H \otimes F$ ,  $G[i,j] = \sum_{u,v} H[u,v]F[i+u,j+v]$ . **Convolution:** kernel flipped,  $(H * F)[i,j] = \sum_{u,v} H[u,v]F[i-u,j-v]$ . For symmetric kernels, correlation = convolution. Boundary: zero fill, replicate border, mirror reflect, mirror across edge. **Linear filters:** shifting, blur, sharpen, edge-like high-pass. Box filter averages but creates blocky response. Gaussian  $G_\sigma(x,y) = \frac{1}{2\pi\sigma^2} \exp(-(x^2+y^2)/(2\sigma^2))$ ; smoothing suppresses high frequency/noise; larger  $\sigma$  means stronger blur. Median filter is nonlinear, robust to salt-and-pepper noise. **Separable filter:**  $K = uv^\top$ , so 2D convolution becomes horizontal then vertical; cost  $O(k^2HW) \rightarrow O(2kHW)$ . Gaussian is separable. Sharpening can be viewed as  $I + \alpha(I - \text{blur}(I))$ . **Kernel facts:** smoothing kernels usually positive and sum to 1, preserving constant images. Output shape choices: full keeps all overlaps; same keeps input size; valid only positions where kernel fully fits. **Convolution properties:** shift-invariant, commutative, associative, distributive, identity impulse  $\delta$ , and  $\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g)$ ; Gaussian blur is low-pass. **Template matching:** filters can be matched templates; use normalized cross-correlation to reduce brightness bias. **Hybrid image:** low-pass one image + high-pass another, perceived content changes with viewing distance/scale. Separable iff  $\text{rank}(K) = 1$ ; SVD gives best rank-1 approximation for nearly separable filters.

## 3 L3 Camera Models

### 3.1 Perspective Projection

Camera center  $C$ , optical axis  $Z$ , image plane distance  $f$ , principal point  $p = (p_x, p_y)$ . For camera coordinate  $Q = (X,Y,Z)^\top$ :

$$x = \frac{fX}{Z} + p_x, \quad y = \frac{fY}{Z} + p_y.$$

Depth ambiguity: one pixel corresponds to a whole ray. Larger focal length  $\Rightarrow$  narrower field of view. **Pinhole/lens:** tiny aperture gives sharp image but little light and diffraction; large aperture gives more light but blur; lens focuses rays from one depth plane onto sensor. Projection loses depth, metric distances and angles; it preserves incidence of points/lines, but objects through camera center may degenerate. **Homogeneous coordinates:**  $(x,y,1)^\top \sim (sx,sy,s)^\top$ . Line  $l = (a,b,c)^\top$  means  $l^\top x = 0$ ; line through two points  $l = x_1 \times x_2$ . Intrinsic:

$$K = \begin{bmatrix} f_x & s & p_x \\ 0 & f_y & p_y \\ 0 & 0 & 1 \end{bmatrix}, \quad \tilde{x} \sim KX_C.$$

Square/no skew:  $s = 0, f_x = f_y = f$ . Full projection:

$$X_C = RX_W + t = R(X_W - C), \quad P = K[R \mid t], \quad \tilde{x} \sim PX_W.$$

After  $\tilde{x} = PX$ , divide by third coordinate.

### 3.2 Projective Properties

Points map to points; 3D lines generally map to image lines; parallel 3D lines may meet at vanishing point. For line  $X(t) = V + tD$ , as  $t \rightarrow \infty$ , projected point depends only on direction  $D$ :

$$\tilde{x}_v \sim KD, \quad D \sim K^{-1}\tilde{x}_v.$$

Vanishing points of directions in a plane lie on the *vanishing line*; ground-plane vanishing line is horizon. Parallel planes share horizon line. Orthographic projection drops depth:

$$(X,Y,Z,1)^\top \mapsto (X,Y,1)^\top$$

with no perspective division; good when depth variation is small relative to distance. **VP caveats:** lines parallel to image plane meet at infinity; 2D image intersection does not imply 3D parallel lines. **Calibration:** known 3D calibration pattern + detected 2D corners estimates  $K$  and extrinsics; three orthogonal vanishing points can calibrate Manhattan scenes. **Single-view metrology:** with flat ground and level camera, horizon height can reveal camera height on vertical objects; with known  $K$  and ground plane, backproject image rays to plane and place CAD/measure scene.

## 4 L4 Stereo & Epipolar Geometry

### 4.1 Parallel Calibrated Stereo

Two views turn a left pixel ray into an epipolar line in right image; matching + triangulation recovers 3D. For rectified parallel cameras with baseline  $B$ , focal  $f$ :

$$y_l = y_r, \quad d = x_l - x_r, \quad Z = \frac{fB}{d}.$$

Closer objects have larger disparity. Full 3D from left image:

$$X = \frac{(x_l - p_x)Z}{f}, \quad Y = \frac{(y_l - p_y)Z}{f}.$$

Patch matching cost for disparity  $d$ :

$$\text{SSD}(p,d) = \sum_{q \in W} (I_l(p+q) - I_r(p - (d,0) + q))^2.$$

Normalized correlation robust to intensity shifts:

$$\text{NC} = \frac{\sum(I_l - \bar{I}_l)(I_r - \bar{I}_r)}{\sqrt{\sum(I_l - \bar{I}_l)^2} \sqrt{\sum(I_r - \bar{I}_r)^2}}.$$

Patch size: small = detailed/noisy; large = stable/blurred boundaries. Smooth stereo energy:

$$E(d) = \sum_p D_p(d_p) + \lambda \sum_{(p,q) \in N} V(d_p, d_q).$$

### 4.2 General Epipolar Geometry

Epipole  $e_r$ : left camera center projected into right image. Epipolar plane  $(C_l, C_r, P)$  intersects images in paired epipolar lines; matching search becomes 1D. Fundamental matrix maps point to line:

$$l_r = Fp_l, \quad p_r^\top Fp_l = 0.$$

Construction:  $F = [e_r]_\times H_\pi$ , where  $H_\pi$  is plane-induced homography and

$$[e]_\times = \begin{bmatrix} 0 & -e_3 & e_2 \\ e_3 & 0 & -e_1 \\ -e_2 & e_1 & 0 \end{bmatrix}.$$

Each correspondence gives linear equation in  $F$  entries:

$$\begin{aligned} x_r x_l f_{11} + x_r y_l f_{12} + x_r f_{13} + y_r x_l f_{21} + y_r y_l f_{22} \\ + y_r f_{23} + x_l f_{31} + y_l f_{32} + f_{33} = 0. \end{aligned}$$

$F$  has scale ambiguity and rank constraint; use multiple matches + least squares/RANSAC. Rectification warps images so epipolar lines become horizontal. From  $F$  one projective camera choice:  $P_l = [I \mid 0]$ ,  $P_r = [[e_r]_\times F \mid e_r]$ . **F facts:**  $F$  has scale ambiguity + rank-2 constraint, so 7 DoF;  $e_r^\top F = 0$  and  $F e_l = 0$ . Need  $n \geq 7$  matches in principle, more with RANSAC in practice. **SfM:** multi-view stereo jointly estimates cameras and 3D points by minimizing reprojection error  $\sum_{ij} \|x_{ij} - \pi(P_i X_j)\|^2$ ; solved by bundle adjustment. **Structured light:** projector emits known pattern to simplify correspondence; Kinect uses IR pattern/depth sensing.

## 5 L5 Keypoints, Features, Matching

**Local feature goal:** match corresponding image points from same world point. Recall needs invariance to rotation/scale/affine/illumination; precision needs contrast/distinctiveness and locality. Raw patch descriptor is sensitive to shifts/rotations. **Detector desiderata:** salient + repeatable under view-point/illumination. Matching all pixels increases false positives unless descriptor is extremely strong.

## 5.1 SIFT & Keypoint Selection

**SIFT Descriptor:** Scale-Invariant Feature Transform. Scale invariance from Gaussian pyramid/scale-space; illumination robustness from normalized gradients; rotation invariance from dominant orientation. Descriptor: rotate local grid to dominant gradient, split into  $4 \times 4$  cells, each with 8-bin gradient histogram  $\Rightarrow$  128D descriptor. **SIFT matching:** nearest neighbor in Euclidean descriptor space; threshold ratio of nearest to 2nd nearest descriptor. Low ratio = good match; high ratio = ambiguous. **Corners:** intensity changes in all directions. Harris uses local second-moment matrix:

$$\Delta^2 \approx \begin{bmatrix} \Delta x & \Delta y \\ D_x D_y & D_y^2 \end{bmatrix} \begin{bmatrix} D_x^2 & D_x D_y \\ D_x D_y & D_y^2 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}.$$

Large eigenvalues in both directions  $\Rightarrow$  corner; one large one small  $\Rightarrow$  edge; both small  $\Rightarrow$  flat. Harris response often  $R = \det M - k(\text{tr } M)^2$ . Scale-invariant interest points: Laplacian of Gaussian approximated by Difference of Gaussian.

### 5.2 Homography & RANSAC

Transformations: translation/rotation/similarity/affine/projective. Affine preserves parallel lines; projective/homography may send parallel lines to vanishing point. Planar scene or pure camera rotation gives homography:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \sim H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \quad H = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & 1 \end{bmatrix}.$$

$H$  has 8 DoF up to scale; each match gives 2 equations;  $\geq 4$  matches, over-constrained solve least squares. Image mosaic: reproject images from same center of projection into a shared synthetic wide-angle camera. **2D transform hierarchy:** translation 2 DoF; similarity 4 DoF (translation+rotation+uniform scale); affine 6 DoF, preserves parallel lines; homography/projective 8 DoF, preserves straight lines but not parallelism. **Warping:** forward warp can create holes/non-integer hits; inverse sampling with interpolation is usually cleaner. Mosaic pipeline: detect/match features  $\rightarrow$  estimate  $H$  with RANSAC  $\rightarrow$  warp to common frame  $\rightarrow$  blend. **RANSAC:** sample minimal set, fit model, count inliers under threshold, repeat, refit on inliers. Iteration count for success probability  $p$ , inlier ratio  $w$ , sample size  $s$ :

$$N = \frac{\log(1-p)}{\log(1-w^s)}.$$

Pros: simple, works with outliers; cons: poor with low inlier ratio, bad thresholds, degenerate samples.

## 6 L6 Vision Neural Networks

### 6.1 CNN Basics

CNN biases: local connectivity, weight sharing, spatial structure. Conv filter  $F \times F \times C$  slides over input; output depth = number of filters  $K$ .

$$W_2 = \frac{W_1 - F + 2P}{S} + 1, \quad H_2 = \frac{H_1 - F + 2P}{S} + 1, \\ \# \theta = F^2 CK + K.$$

Pooling: per-channel downsample, no parameters; max pooling  $W_2 = (W_1 - F)/S + 1$ .  $1 \times 1$  conv mixes channels only, useful for projection/bottleneck. ReLU  $f(x) = \max(0,x)$ : efficient, non-saturating positive side, but dead ReLU/output not zero-centered. **FC vs Conv:** FC flattens  $HWC$  into a vector; params for  $M$  outputs are  $HWC \cdot M + M$ , no spatial sharing. Conv keeps the grid, uses local receptive fields and shared weights. Sigmoid/tanh saturate, kill gradients, are not zero-centered (sigmoid), and use expensive exp; ReLU is fast/non-saturating for  $x > 0$  but zero-gradient for  $x < 0$ ; LeakyReLU/ELU mitigate dead units. **Initialization/normalization:** constant init breaks symmetry poorly; small random can vanish/explode. Xavier std  $1/\sqrt{D_{in}}$ . LayerNorm:

$$\mu = \frac{1}{d} \sum_j x_j, \quad \sigma^2 = \frac{1}{d} \sum_j (x_j - \mu)^2, \quad y_j = \gamma \frac{x_j - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta.$$

Regularization/generalization: early stopping, data augmentation, dropout, stochastic depth, random crop, Mixup. **Regularization view:** always keep best validation snapshot; augmentation/dropout/stochastic depth can be read as training with injected noise, while test-time prediction marginalizes/averages over the noise.

## 6.2 Architectures

AlexNet: ReLU, data augmentation, GPU split, ImageNet 2012. ResNet solves deep optimization by residual block  $H(x) = F(x) + x$ ; if  $F = 0$  identity is easy. Bottleneck:  $1 \times 1$  reduce,  $3 \times 3$ ,  $1 \times 1$  expand. DenseNet: each layer connects to all later layers, feature reuse, gradient flow. Depthwise separable conv decomposes standard conv into depthwise spatial + pointwise channel:

$$K^2C^2 \rightarrow K^2C + C^2.$$

ConvNeXt: Transformer ideas in CNN: patchify stem, depthwise conv, inverted bottleneck, GeLU, BatchNorm→LayerNorm, fewer norm layers, separate downsampling. U-Net: encoder-decoder with long skip connections for localization; descendants in segmentation, diffusion, point cloud. **ResNet head**: drops large FC stack; last conv → global average pooling → class FC, reducing params and supporting variable spatial input. **ConvNeXt V2**: not just architecture; ConvNeXt blocks are co-designed with MAE-style/self-supervised training.

## 6.3 Attention, Transformer, ViT

Attention with query  $Q$ , data  $X$ :  $K = XW_K, V = XW_V$ ,

$$E = \frac{QK^\top}{\sqrt{D}}, \quad A = \text{softmax}(E), \quad Y = AV.$$

Self-attention:  $Q = XW_Q, K = XW_K, V = XW_V$ . Masked attention sets illegal future logits to  $-\infty$ . Multi-head:  $H$  heads,  $D_H = D/H$ , concatenate then output projection. Complexity/memory  $O(N^2)$  due to attention matrix; FlashAttention avoids materializing full  $N \times N$  matrix. Transformer block: self-attention is vector interaction; LayerNorm/MLP are per-token; pre-norm makes residual identity easier. ViT: split image into patches, flatten, linear project to tokens, add positional embedding, Transformer, pool/[CLS], classify. Example  $16 \times 16 \times 3 = 768$  patch vector. ViT weakens CNN translation equivariance but gives global interaction. RMSNorm:

$$y_i = \frac{x_i}{\sqrt{e + \frac{1}{N} \sum_i x_i^2}} \gamma_i.$$

# 7 L7 Classification & Representation Learning

## 7.1 Recognition & Self-Supervision

Recognition tasks: identification, scene/object classification, annotation, detection, segmentation, pose, attributes, action. Classification challenge: semantic gap, viewpoint, clutter, illumination, occlusion, deformation, intra-class variation. Supervised pipeline: collect labels, train classifier, evaluate; expensive labels motivate self-supervised learning (SSL). **Pretext vs downstream**: labels generated from data itself for pretraining; downstream is real labeled task. **Pretext risk**: auto-generated labels may teach task artifacts rather than transferable semantics if the pretext task is too narrow. **MAE**: mask high ratio patches (e.g. 75%), encoder sees visible patches only, decoder reconstructs pixels at masked patches; keep encoder, discard decoder. Loss:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|x_i - \hat{x}_i\|_2.$$

Asymmetric: large encoder, small decoder.

## 7.2 Contrastive Learning

Goal: positive pairs close, negatives far. InfoNCE:

$$L = -\mathbb{E} \log \frac{\exp(s(f(x), f(x^+)))}{\exp(s(f(x), f(x^+))) + \sum_{j=1}^{N-1} \exp(s(f(x), f(x_j^-)))}.$$

SimCLR: two augmentations of same image form positive; cosine score  $s(u, v) = u^\top v / (\|u\| \|v\|)$ ; encoder  $h = f(\tilde{x})$ , projection head  $z = g(h)$ . NT-Xent for positive  $(i, j)$ :

$$\ell_{ij} = -\log \frac{\exp(s(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(s(z_i, z_k)/\tau)}.$$

Downstream uses  $h$  not  $z$ . MoCo: queue of keys decouples negatives from batch size; key encoder EMA:

$$\theta_k \leftarrow m\theta_k + (1 - m)\theta_{q_i}.$$

DINO: student matches teacher outputs; teacher EMA, centering+sharpening avoid collapse:

$$\theta_t \leftarrow \tau\theta_t + (1 - \tau)\theta_s.$$

DINOv2 scales curated data + global DINO + local masked/iBOT, strong patch features. **SSL eval**: pretrain encoder, discard projection head, then freeze+linear probe or fine-tune on downstream labels.

## 7.3 JEPA & Vision-Language

JEPA aligns embeddings  $s_x = \text{Enc}_x(x), s_y = \text{Enc}_y(y)$ . JEPA predicts target representation:

$$\hat{s}_y = \text{Pred}(\text{Enc}_x(x), z), \quad \mathcal{L} = \|s_y - \hat{s}_y\|^2.$$

Not pixel reconstruction; predicts abstract representation and avoids irrelevant texture. Because  $\text{Enc}_y$  is not invertible, prediction in representation space need not reconstruct full  $y$ . H-JEPA passes latents hierarchically to predict more abstract/longer-term structure. Collapse: all inputs map to constant; SIGReg regularizes latent statistics to avoid collapse. **CLIP**: image-text contrastive pretraining on pairs, open vocabulary. Batch similarity:

$$I_i = f_{\text{img}}(x_i), \quad T_j = f_{\text{text}}(t_j), \quad S_{ij} = I_i^\top T_j.$$

Diagonal matched, off-diagonal negatives; train image-to-text and text-to-image. Zero-shot:

$$\hat{c} = \arg \max_{c \in \mathcal{C}} f_{\text{img}}(x)^\top f_{\text{text}}(\text{"a photo of a c"}).$$

SigLIP replaces softmax contrastive with sigmoid pairwise loss, less batch-size coupled. CoCa combines CLIP-style contrastive with captioning decoder. **VLM notes**: CLIP has modality gap; image-level captions give weak local/region supervision, so slides discuss region captions and VLM training. LLaVA connects visual encoder to LLM; Flamingo uses gated cross-attention, masked attention, and in-context multimodal learning.

# 8 L8 Object Detection

## 8.1 Task & Metrics

Detection output set  $\{(b_i, c_i, s_i)\}_{i=1}^N$ : boxes, class, confidence. Harder than classification: variable count/scale/location, overlaps, background. Approaches: sliding window, region proposal, implicit shape/codebook, modern learned queries. **Implicit Shape Model**: local features match codebook entries; each match casts vote for object center/scale, Hough-style detection and rough segmentation. **IoU**:  $\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}$ . Precision/recall:

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}.$$

AP = area under PR curve after sorting detections by confidence; mAP = mean over classes/IoU thresholds. Each GT matched once; duplicate boxes become FP. Anchor/proposal box regression often uses

$$t_x = \frac{x - x_a}{w_a}, \quad t_y = \frac{y - y_a}{h_a}, \quad t_w = \log \frac{w}{w_a}, \quad t_h = \log \frac{h}{h_a}.$$

Detection loss:

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda \mathcal{L}_{box}.$$

## 8.2 R-CNN Family

R-CNN: Selective Search  $\sim 2000$  proposals, crop/warp each, CNN per proposal, classify + box correction; slow due to repeated convolution. Fast R-CNN: one whole-image backbone, crop shared feature with RoIPool, heads. RoIPool maps variable RoI to fixed  $7 \times 7$  by quantized bins; quantization hurts alignment. Faster R-CNN: learn proposals with RPN. For feature map  $H \times W$  and  $K$  anchors/location: candidates  $KHW$ , objectness logits  $2KHW$ , box offsets  $4KHW$ ; sort by objectness and keep top  $\sim 300$  proposals. Two-stage = RPN proposals then RoI classify/regress. **Single-stage**: YOLO/SSD/RetinaNet dense predict on grid. YOLO output:

$$S \times S \times (5B + C),$$

where each box has objectness +  $(x, y, w, h)$  and classes. NMS: keep highest score, remove lower boxes with high IoU, repeat. Final YOLO class-specific score is roughly  $P(\text{obj})P(c)$ ; cell/grid assignment makes dense detection fast but can struggle with crowded small objects.

## 8.3 DETR & Open Vocabulary

DETR: CNN/ViT features + positional encoding + Transformer encoder + learned object queries in decoder; each query predicts class (incl. no-object) and box. Set prediction uses Hungarian matching:

$$\hat{\sigma} = \arg \min_{\sigma} \sum_{j=1}^M \mathcal{L}(y_j, \hat{y}_{\sigma(j)}),$$

cost = class + L1 box + GloU. Matching encourages one prediction per GT, reducing NMS need. Tradeoff: clean end-to-end but slow convergence/small objects. Open-vocabulary detection replaces fixed class head by text queries. OWL-ViT: CLIP-style pretraining then detection fine-tuning; removes CLIP pooling so per-patch/token image embeddings compare to text/image queries, while a box MLP predicts coordinates. Text embedding acts as classifier weights. OWLv2/self-training expands detection supervision. Grounding DINO: language-aware open-set detector; bridges text phrase to boxes.

# 9 L9 Dense Prediction

## 9.1 Segmentation, FCN, Upsampling

Dense prediction:  $I \in \mathbb{R}^{H \times W \times 3} \rightarrow Y \in \mathbb{R}^{H \times W \times C}$ . Semantic segmentation = class per pixel; instance = distinguish object instances; panoptic = stuff + things. Depth  $D \in \mathbb{R}^{H \times W}$ ; flow  $C = 2$ ; diffusion predicts dense noise/clean image. Core tension: semantic abstraction vs spatial precision. FCN converts classifier to fully convolutional:  $1 \times 1$  conv over feature map gives low-res class score map; bilinear or learned transposed-conv upsampling restores resolution. Skip fusion combines high-level semantic coarse maps with low-level fine maps; FCN fuses skip logits by summing. Segmentation metrics:

$$\text{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c}, \quad mIoU = \frac{1}{C} \sum_c \text{IoU}_c.$$

Upsampling: interpolation, transposed convolution (linear operator transpose, not inverse), max unpooling with switches, resize+conv to reduce checkerboard. Stride-2 transposed conv can be viewed as inserting zero rows/cols then convolving with flipped kernel; uneven overlap causes checkerboard. DeconvNet: encoder-decoder; U-Net concatenates encoder features with decoder features and predicts at the end; FPN: top-down + lateral connections to semantic pyramid.

## 9.2 Mask R-CNN, SAM, Depth, Generation

Mask R-CNN = Faster R-CNN + FCN mask head on Rols. Outputs  $\{(b_i, c_i, m_i)\}$ . Mask head  $M \in [0, 1]^{K \times m \times m}$ , uses per-pixel sigmoid BCE for GT class. RoAlign avoids RoIPool quantization using bilinear sampling; crucial for masks/keypoints. Keypoints as heatmaps  $\hat{H} \in \mathbb{R}^{K \times H \times W}$ . SAM: promptable segmentation foundation model. Components: heavy ViT image encoder (cacheable), prompt encoder (points/boxes/text/masks), lightweight mask decoder with two-way attention. Outputs multiple masks for ambiguity + predicted IoU for ranking. Limitations: may miss fine structures, hallucinate small disconnected components, fail boundaries/text-to-mask, and lose to domain-specific tools. SA-1B built with model-in-the-loop stages; SAM 2 stores previous-frame memory for video, preserving object identity under motion/occlusion and supporting streaming interaction. Depth Anything: teacher/student, labeled + 65M unlabeled pseudo labels, DINOv2 backbone; perturbation consistency (color jitter/blur/CutMix) lets student generalize beyond teacher. V2 uses synthetic data for precise depth then bridges sim-to-real. Single-image depth is not true 3D and not necessarily multi-view consistent. Generation as dense prediction: GAN/cGAN/Pix2Pix/DDPM/LDM/DiT all predict dense maps over pixels/latents/patches. PatchGAN outputs spatial realism map over local patches; LDM denoises latent  $z_i$ ; DiT replaces U-Net denoiser with ViT.

# 10 L10 Synthetic Data

**Why**: controllable labels hard to collect in real world: depth, normal, optical flow, scene flow, long-term tracking, 6D pose. Standard paradigm: pretrain on synthetic, adapt/fine-tune/evaluate on real. **Domain gap**: synthetic accuracy  $\neq$  real accuracy; gap comes from rendering, texture, lighting, object distribution, sensor noise, behavior. **Generation paradigms**: 3D/game engines are realistic and physically grounded but asset-limited; procedural generation gives infinite scenes + perfect ground truth but may look synthetic; GenAI looks realistic but geometry/labels may be noisy. Key sources: FlyingThings3D/SceneFlow, Kubric/MOVI, Hypersim, Replica, GTA5/SYNTHIA/AI2-THOR, Virtual KITTI.

## 10.1 Tracking & Pose

Synthetic tracking provides full trajectories and occlusion labels. Point tracking task: given query point in a frame, predict its location and visibility across time; long-term trajectories and occlusion flags are almost impossible to annotate densely by hand. Kubric uses Blender+PyBullet to produce RGB/depth/normal/segmentation/flow/trajectories/visibility. PIPs: persistent independent particles track any point; TAP-Vid-Kubric benchmark; TAP-Net/TAPIR refine per-frame matching into robust point tracking; CoTracker tracks points jointly with cross-track attention; BootsTAPIR/CoTracker3 use teacher-student pseudo labels; Track-On-R adds verifier-guided filtering. 6-DoF pose: model-based uses CAD model/rendering; model-free/few-shot matches RGB-D/reference observations. FS6D uses few-shot ShapeNet6D, online augmentation (texture blending/deformation). MegaPose: render-and-compare. FoundationPose: unified framework trained purely synthetic. Metrics:

$$ADD = \frac{1}{|M|} \sum_{x \in M} \|Rx + t - (R^*x + t^*)\|,$$

$$ADD-S = \frac{1}{|M|} \sum_{x_1 \in M} \min_{x_2 \in M} \|Rx_1 + t - (R^*x_2 + t^*)\|$$

for symmetric objects.

## 10.2 Depth, Normal, Sensors

Marigold fine-tunes Stable Diffusion for depth; DAC generalizes to arbitrary cameras incl. ERP panorama. Metric depth is absolute meters; relative/affine-invariant depth preserves ordering/shape up to scale/shift. DAC maps arbitrary FoV cameras into an ERP universal canvas via FoV alignment, then maps prediction back. Camera Depth Models (CDM) denoise RGB + raw noisy sensor depth into clean metric depth, not monocular depth. Surface normal: Omnidata uses 3D scans for mid-level cues; normals are ill-posed from images due to shading/texture/lighting ambiguity. DSINE injects geometry bias: ray direction encoding from intrinsics, Ray ReLU visible hemisphere, local normal rotation, iterative refinement. StableNormal uses diffusion prior with YOSO init + SG-DRN refinement. Key lesson: right inductive bias can replace huge data scale.

## 11 L11 Curating Vision-Language Data

**Alt-text web data:** large scale but noisy/mismatched. Uses: generation, VLM, VLA. Pipeline: crawl image-text, text filtering, image-text alignment verification, dedup/safety/balancing. **Three schools:** text-only filtering (WIT/MetaCLIP), image-text alignment verification (CC3M/CC12M/LAION/DataComp/DFN), and embracing noise/scale (ALIGN/No Filter/WebLI). Quality controls precision; diversity controls style/long-tail/cultural coverage; safety filtering is still needed even when semantic filtering is relaxed.

### 11.1 Text-Side Filtering

General: build vocabulary/concepts, match text, balance counts. WIT behind CLIP; WIT vs YFCC: scale and web alt-text matter. MetaCLIP demystifies CLIP data: use metadata vocabulary; cap per concept (e.g. 20k) to balance frequent/rare terms; performance comes from transparent curation + scale.

### 11.2 Image-Text Alignment

Filtering paradox: stricter filters remove noise but also diversity/long tail. CC3M: harvest alt-text, image filtering, text filtering, Cloud Vision API tag overlap, hypernymization; high precision but low recall. CC12M keeps image-text verification but relaxes unimodal filters, increasing scale/diversity with lower precision. LAION uses CLIP cosine threshold and releases URLs+metadata; democratizes scale but inherits CLIP filter bias and lacks concept balancing. DataComp: benchmark dataset curation strategies under fixed compute; top-30% CLIP score is a Western-benchmark sweet spot, not universal optimum; dedup improves generalization. DFN trains a dedicated filter on high-quality image-text pairs; filter quality  $\neq$  ImageNet accuracy, and poisoned/low-quality filter training data hurts badly. No Filter/WebLI: at 10B–100B scale, random noise can dilute but systematic bias/stereotypes amplify; global pretraining + short English fine-tune preserves diversity. Scale helps coverage parity and multilingual/cultural performance, not automatic stereotype correction. Datasets still need NSFW/toxicity/PII/watermark/low-quality filtering, but overfiltering harms coverage.

## 12 L12 Diffusion & Score Models

### 12.1 EBM, Score, DSM

Density modeling wants sample from  $p(x)$ . EBM:  $p_\theta(x) = \exp(-E_\theta(x))/Z_\theta$ ,  $Z$  intractable. Langevin dynamics uses score:

$$x_{k+1} = x_k + \frac{\eta}{2} \nabla_x \log p(x_k) + \sqrt{\eta} \epsilon_k.$$

Score matching learns  $s_\theta(x) \approx \nabla_x \log p(x)$  without  $Z$ :

$$\begin{aligned} \mathcal{J}_{\text{SM}} &= \frac{1}{2} \mathbb{E}_p \|\mathbf{s}_\theta(x) - \nabla_x \log p(x)\|^2 \\ &\equiv \mathbb{E}_p \left[ \frac{1}{2} \|\mathbf{s}_\theta(x)\|^2 + \nabla_x \cdot \mathbf{s}_\theta(x) \right] + \text{const.} \end{aligned}$$

Hyvarinen score matching removes unknown  $\nabla \log p$  by integration by parts; denoising score matching avoids second derivatives by learning scores of noisy conditional Gaussians. With  $x_{\mathcal{O}} = x + \sigma \epsilon$ :

$$\nabla_{x_{\mathcal{O}}} \log p_{\mathcal{O}}(x_{\mathcal{O}}|x) = -(x_{\mathcal{O}} - x)/\sigma^2.$$

NCSN learns scores at multiple  $\sigma$ ; annealed Langevin samples from large noise to small noise. Metropolis-Hastings accepts/rejects random-walk proposals but mixes slowly in high dimensions; Langevin uses the score direction for informed updates. Annealed Langevin: for  $\sigma_1 > \dots > \sigma_L$ , run several Langevin steps at each noise level, coarse-to-fine.

### 12.2 DDPM

Forward:

$$q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I), \quad \alpha_t = 1 - \beta_t,$$

$$\tilde{\alpha}_t = \prod_{s=1}^t \alpha_s, \quad q(x_t|x_0) = \mathcal{N}(\sqrt{\tilde{\alpha}_t}x_0, (1 - \tilde{\alpha}_t)I),$$

$$x_t = \sqrt{\tilde{\alpha}_t}x_0 + \sqrt{1 - \tilde{\alpha}_t}\epsilon.$$

Reverse:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(\mu_\theta(x_t, t), \Sigma_\theta(x_t, t)).$$

Posterior:

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(\tilde{\mu}_t, \tilde{\beta}_t I),$$

$$\tilde{\mu}_t = \frac{\sqrt{\tilde{\alpha}_{t-1}}\beta_t}{1 - \tilde{\alpha}_t} x_0 + \frac{\sqrt{\tilde{\alpha}_t}(1 - \tilde{\alpha}_{t-1})}{1 - \tilde{\alpha}_t} x_t, \quad \tilde{\beta}_t = \frac{1 - \tilde{\alpha}_{t-1}}{1 - \tilde{\alpha}_t} \beta_t.$$

Noise parameterization:

$$x_0 = \frac{x_t - \sqrt{1 - \tilde{\alpha}_t}\epsilon}{\sqrt{\tilde{\alpha}_t}}, \quad \mathcal{L}_{\text{simple}} = \mathbb{E}_{t, x_0, \epsilon} \|\epsilon - \epsilon_\theta(x_t, t)\|_2^2.$$

Sampling:

$$x_{t-1} = \frac{1}{\sqrt{\tilde{\alpha}_t}} \left( x_t - \frac{\beta_t}{\sqrt{1 - \tilde{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z.$$

### 12.3 Continuous View & Equivalent Targets

SDE/PF-ODE unify score/diffusion:

$$\begin{aligned} dx &= f(x, t) dt + g(t) dw, \\ dx &= \left[ f(x, t) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(x) \right] dt. \end{aligned}$$

Gaussian path:

$$z_t = \alpha_t x + \sigma_t \epsilon, \quad \text{SNR} = \alpha_t^2 / \sigma_t^2, \quad \lambda = \log \text{SNR}.$$

Equivalent targets: clean  $x$ , noise  $\epsilon$ , score  $s_t(z) = -\mathbb{E}[\epsilon|z_t = z]/\sigma_t$ , velocity/path derivative  $\dot{v}_t = \dot{\alpha}_t x + \dot{\sigma}_t \epsilon$ , denoiser  $D(z_t) = \mathbb{E}[x|z_t]$ . Weighting and schedule can be expressed in SNR/log-SNR coordinates; epsilon-prediction is not uniform in SNR, log-SNR makes interpretation cleaner. **Forward paths:** VE/EDM  $x_t = x_0 + \sigma_t \epsilon$ ; VP/DDPM  $x_t = \alpha_t x_0 + \sigma_t \epsilon$ ,  $\alpha_t^2 + \sigma_t^2 = 1$ ; Flow Matching  $x_t = (1 - t)x_0 + t\epsilon$ . Only VP keeps variance normalized. **VLB weight:** in  $x_0$  prediction, DDPM VLB weights each step by SNR drop,

$$L_{t-1} \propto (r_{t-1} - r_t) \|D_\theta(x_t, t) - x_0\|^2, \quad r = \text{SNR}.$$

**Training design:** choose path, prediction target  $(\epsilon, x_0, s, v)$ , objective weight  $w(\lambda)$ , and noise proposal/schedule  $q(\lambda)$ ; Monte Carlo estimator uses weight  $w/q$ . Slides compare uniform, EDM/sigmoid, and Min-SNR weighting; monotone weighted objectives can be rewritten as mixtures of ELBOs on Gaussian-noise-augmented data.

## 13 L13 Beyond Diffusion

### 13.1 Normalizing Flow

Flow uses invertible  $z = f_\theta(x)$  with exact likelihood:

$$\log p_X(x) = \log p_Z(f_\theta(x)) + \log \left| \det \frac{\partial f_\theta(x)}{\partial x} \right|.$$

MLE minimizes negative log-likelihood. Composition: log-det sums over layers. Bottleneck: invertibility + tractable Jacobian determinant, dimension match. Triangular Jacobian makes determinant product of diagonals. NICE additive coupling:

$$y_a = x_a, \quad y_b = x_b + m(x_a), \quad \det J = 1.$$

RealNVP affine coupling:

$$y_a = x_a, \quad y_b = x_b \odot \exp(s(x_a)) + t(x_a), \quad \log |\det J| = \sum s(x_a).$$

TarFlow: transformer masked autoregressive flow over patches; triangular Jacobian via ordering. Noise augmentation + Tweedie denoising:

$$\mathbb{E}[x|y] = y + \sigma^2 \nabla_y \log p(y).$$

Guidance analogous classifier-free: combine conditional/unconditional scores/log-probs.

## 13.2 GAN, AR, Masked/Flexible-Order

GAN objective:

$$\min_G \max_D \mathbb{E}_{x \sim p_{data}} \log D(x) + \mathbb{E}_{z \sim p_z} \log(1 - D(G(z))).$$

Pros: fast high-fidelity samples; cons: unstable, mode collapse, no exact likelihood. Optimal discriminator:

$$D^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_G(x)},$$

$$V(D^*, G) = -\log 4 + 2 \text{JSD}(p_{data} \| p_G).$$

If supports barely overlap, JS can saturate and generator gradients become poor. WGAN replaces JS by Earth-Mover/Wasserstein-1:

$$W_1(p, q) = \sup_{\|f\|_L \leq 1} \mathbb{E}_p f(x) - \mathbb{E}_q f(x),$$

with 1-Lipschitz critic, giving useful gradients even for non-overlapping supports. RpGAN compares paired real/fake scores; R3GAN adds principled R1/R2-style regularization to stabilize dynamics. Modern GANs also survive as perceptual/adversarial losses for diffusion, distillation, and few-step generators. AR factorizes exact likelihood:

$$p(x) = \prod_{i=1}^N p(x_i | x_{<i}),$$

stable MLE/cross-entropy but serial  $O(N)$  sampling and ordering bias. Image AR tokenizes pixels/patches/codebook tokens; Transformer provides long-range context. **AR evolution:** PixelRNN models raw pixels but training/inference are sequential; PixelCNN uses masked convolution so training is parallel over pixels but sampling remains sequential. VQ-VAE shortens image sequence via discrete codebook tokens. Loss has reconstruction + codebook + commitment terms; straight-through estimator copies gradient through nearest-code lookup. JetFormer-style soft tokens use invertible flow tokenizer + AR Transformer, preserving continuous information and exact likelihood more directly than hard codebooks. Masked/flexible-order generation predicts masked tokens in iterative refinement; faster than AR, useful for editing/inpainting, likelihood often approximate or schedule-dependent. **MaskGIT:** VQ-GAN tokenizer + bidirectional Transformer; train by masking random token subset and cross-entropy on masked tokens only. Iterative decoding predicts all masked positions in parallel, samples tokens, keeps high-confidence ones, and follows a mask schedule; steps drop from  $O(N)$  to about 8–12, but bidirectional attention cannot use KV cache. **RandAR:** random-order causal generation adds position instruction tokens, allowing KV-cache and parallel decoding of multiple requested positions while avoiding fixed raster order. **Comparison:** Flow exact likelihood/fast/stable but Jacobian constraints; GAN no likelihood/fast but unstable and mode collapse; AR exact likelihood/stable but slow serial order; Masked approx likelihood/medium speed with mask schedule; Diffusion ELBO/score/stable but many denoise steps.